

Bhavik Kumar

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Quantum computation and information | Quantum simulation of many-body physics

EDUCATION

University Of Copenhagen

PhD fellow, Theoretical and Mathematical Physics (QMATH)

Supervision: Prof. Daniel Malz

Copenhagen, Denmark

September 2023 – Present

Indian Institute of Science Education and Research, Mohali

Integrated BS-MS physics (Dual Degree)

Mohali, India

August 2018 – June 2023

RESEARCH OBJECTIVES

My current research interests are centered around Quantum Computation and information and its application in simulation/control/communication of quantum systems. I am passionate about investigating various aspects of fundamental physics and its applications, especially complex quantum many-body phenomena, by harnessing the potential of synthetic quantum matter platforms and quantum technologies.

RESEARCH EXPERIENCE

University of Copenhagen

PhD fellow, Theoretical and Mathematical Physics

Copenhagen, Denmark

September 2023 – Present

- **Topic** – Learning from Dynamics in Analog Quantum Simulators.
- Developed Hamiltonian-driven shadow tomography protocol for the characterization of analogue quantum simulators.
- Developed calibration methods for estimating applied potentials in cold-atom lattice simulators from quench dynamics using tools from Hamiltonian-learning.

University of Cambridge

S.t John's College and Cavendish Laboratory — Visiting PhD student

Cambridge, United Kingdom

May 2025 – July 2025

- **Topics** – Quantum dynamics in dual unitary circuits.
- **Supervisors** - Dr. Max McGinley (TCM group, Dept. of physics) and Prof. Bruno Bertini (University Of Birmingham)
- Investigated hierarchical generalizations of dual unitary quantum circuits, formulating stabilizer-based conditions using clifford algebra.
- Working on developing circuit-level diagnostics of solvability via spatio temporal correlation functions and operator space entanglement for non-clifford circuits.

Ludwig Maximilian University of Munich

External Master's thesis

Munich, Germany

Jan 2022 – April 2023

- **Title** – Stabilizing Quantum Simulators of Gauge Theories against $1/f$ Noise.
- **Grade** – A (Highest possible grade)
- **External Supervisors** – Prof. Jad C. Halimeh (MCQST, LMU Munich) and Prof. Philipp Hauke (University of Trento)
- **Local Supervisor** – Prof. Manabendra Nath Bera (IISER Mohali)
- Illustrated the efficacy of linear gauge protection in stabilizing salient features of gauge theories in quantum simulators, such as gauge invariance against $1/f^\beta$ noise sources.
- Results showed immediate applicability in modern analog quantum simulators and digital NISQ devices, paving the way for further development in the field of quantum simulation of lattice gauge theories

Theoretical Sciences, PPISR

Summer research internship

Bangalore, India

May 2021 – July 2021

- **Title** – Non-markovian dynamics in open quantum systems.
- **Supervision** – Prof R. Srikant (PPISR), Dr. Vinayak Jagdish (University of Krakow, Poland)
- Captured non-markovianity in a quantum channel resulting from mixing Pauli semigroups at the level of intermediate dynamical maps by performing quantum process tomography on IBM's quantum simulators.

INDUSTRY EXPERIENCE

TAQBIT labs PVT LTD

Bangalore, India

Research Intern

May 2020 – Dec 2021

- **Title**–Quantum random number generation and Quantum key distribution.
Supervision– Animesh Aaryan (CEO TAQBIT labs)
- Simulated spontaneous parametric down-conversion in photons to generate quantum entanglement.
- Worked on experimental protocols to extract cryptographically secure random numbers using spatially entangled source quantum random number generator.
- Characterized electro-optic modulators to realize differential phase shift quantum key distribution

PUBLICATIONS AND PREPRINTS

- **Bhavik Kumar**, Daniel Stilck França, Emilio Onorati, Albert Werner, Daniel Malz. *Stochastic Hamiltonian-Driven Local Shadow Tomography*. In preparation.
- **Bhavik Kumar**, Daniel Malz. *Estimating applied potentials in cold-atom lattice simulators*. Phys. Rev. A 113, 033312 (2026) .
- **Bhavik Kumar**, Philipp Hauke, Jad C. Halimeh. *Suppression of $1/f$ noise in quantum simulators of gauge theories*. arXiv:2210.06489.

AWARDS AND SCHOLARSHIPS

- Travel grant from the Department of Mathematical Sciences, University of Copenhagen, for a summer research stay at the Cavendish Laboratory, University of Cambridge (Summer 2025; €4,500).
- Undergraduate and master’s scholarship from the Indian Air Force Benevolent Association (IAFBA).

TEACHING EXPERIENCE

- Teaching Assistant — M.Sc. course *Introduction to Quantum Computing*, Fall 2023–2025, Department of Mathematical Sciences, University of Copenhagen.

SKILLS

- **Software and Programming:** Python (NumPy, SciPy, Matplotlib), quantum simulation and quantum information libraries (QuTiP, QuSpin, Qiskit, Stim), randomized measurements library (RandomMeas.jl), tensor-network frameworks (ITensor, Quimb), Mathematica, MATLAB, PennyLane(Elementary Proficiency), L^AT_EX, Git/GitHub, Visual Studio Code.

SERVICE

- Sub-reviewer for the conference QCTiP 2026.

CONFERENCES/ WORKSHOPS

- SimQuDyn 2026, Mar 29–April 4, LMU Munich, Germany
- QIP 2026, Jan 24–30, Riga, Latvia
- QMATH MasterClass in Representation Theory in Quantum Information science, Aug 18–22, 2025, University of Copenhagen
- QMATH MasterClass in Quantum Simulation, Aug 19–23, 2024, University of Copenhagen
- Geometry and non-adiabatic responses in non-equilibrium systems, June 19–26, 2024, Max Planck Institute for the Physics of Complex Systems, Dresden
- Calbay Quantum School, June 9–14, 2024, Institute for Advanced Study, TU Munich
- Quantum Simulators, Feb 4–9, 2024, Institut Henri Poincaré, Paris
- QMATH MasterClass in Quantum Learning Theory, Aug 19–23, 2023, University of Copenhagen
- Qiskit Global Summer School in Quantum Simulation 2022, Online